Final Project - Code

CCT416 Final Project - Social Data Analytics

Group 5:

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Research Question: "Among university students, which behavioural and demographic characteristics (e.g., sleep hours, social media use, age, diet quality) are most associated with academic performance?"

This project uses Exportratory Data Analysis (EDA), Correlation Analysis, feature importance analysis, and OLS Regression to identify the strongest predictors of academic performance based on a student habits/demographic dataset.

1. Import Libraries

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
import seaborn as sns
import numpy as np
import math
import statsmodels.api as sm
from sklearn.ensemble import RandomForestRegressor
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import MinMaxScaler

# CATEGORICALS_ENCODED is set to True once the categorical variables in the data ar
CATEGORICALS_ENCODED = False
```

We imported all data science libraries we are planning to use for data manipulation, visualization, and regression modelling.

2. Load DataSet

In [2]:
```
df = pd.read_csv('student_habits_performance.csv')
```

We load the dataset 'student_habits_performance.csv', which contains 1000 records on students' lifestyle habits, demographic, and academic performance.

Below is a preview of the first few rows:

In [3]:
```
df.head()
```

Out[3]:

	student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_t
0	S1000	23	Female	0.0	1.2	1.1	
1	S1001	20	Female	6.9	2.8	2.3	
2	S1002	21	Male	1.4	3.1	1.3	
3	S1003	23	Female	1.0	3.9	1.0	
4	S1004	19	Female	5.0	4.4	0.5	

In [4]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 16 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   student_id                    1000 non-null   object
 1   age                           1000 non-null   int64
 2   gender                        1000 non-null   object
 3   study_hours_per_day           1000 non-null   float64
 4   social_media_hours            1000 non-null   float64
 5   netflix_hours                 1000 non-null   float64
 6   part_time_job                 1000 non-null   object
 7   attendance_percentage         1000 non-null   float64
 8   sleep_hours                   1000 non-null   float64
 9   diet_quality                  1000 non-null   object
 10  exercise_frequency            1000 non-null   int64
 11  parental_education_level      909 non-null    object
 12  internet_quality              1000 non-null   object
 13  mental_health_rating          1000 non-null   int64
 14  extracurricular_participation 1000 non-null   object
 15  exam_score                    1000 non-null   float64
dtypes: float64(6), int64(3), object(7)
memory usage: 125.1+ KB
None
```

In [5]:
```
print(df.columns)
```

```
Index(['student_id', 'age', 'gender', 'study_hours_per_day',
       'social_media_hours', 'netflix_hours', 'part_time_job',
       'attendance_percentage', 'sleep_hours', 'diet_quality',
       'exercise_frequency', 'parental_education_level', 'internet_quality',
       'mental_health_rating', 'extracurricular_participation', 'exam_score'],
      dtype='object')
```

3. Data Cleaning and Preprocessing

We check for missing values, convert categorical variables into numeric format, and prepare the dataset for regression model.

3.1 Data Cleaning - Missing (null) values

In [6]:
```
# Check missing values
missing_values = df.isnull().sum()
print("Missing Values Per Column:\n", missing_values)
```

```
Missing Values Per Column:
 student_id                        0
age                               0
gender                            0
study_hours_per_day               0
social_media_hours                0
netflix_hours                     0
part_time_job                     0
attendance_percentage             0
sleep_hours                       0
diet_quality                      0
exercise_frequency                0
parental_education_level         91
internet_quality                  0
mental_health_rating              0
extracurricular_participation     0
exam_score                        0
dtype: int64
```

There are **91 missing values** on **parental_education_level** column. However, after closer inspection of the values in the dataset, it becomes clear that the missing data is actually not missing data. The values that appear as 'None' for parental_education_level indicate the parent has had no education.

This assumption can be made because
(1) the data was generated artifically, made great for training, indicating that missing values are unlikely
(2) the stats displayed in source site (kaggle) indicate that there are no empty entries for this variable.

In [7]:
```
# Handle null values

# Fill missing values in parental_education_level to be "No Education"
df['parental_education_level'].fillna("No Education", inplace=True)
```

```
C:\Users\Public\Documents\ESTsoft\CreatorTemp\ipykernel_27404\2128505126.py:4: Futur
eWarning: A value is trying to be set on a copy of a DataFrame or Series through cha
ined assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because
the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method
({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform
the operation inplace on the original object.


  df['parental_education_level'].fillna("No Education", inplace=True)
```

In [8]:

```python
# Final data check: re-check missing values and data types
print('Missing Values Per Column:\n', df.isnull().sum())

df.info()
```

```
Missing Values Per Column:
 student_id                       0
age                               0
gender                            0
study_hours_per_day               0
social_media_hours                0
netflix_hours                     0
part_time_job                     0
attendance_percentage             0
sleep_hours                       0
diet_quality                      0
exercise_frequency                0
parental_education_level          0
internet_quality                  0
mental_health_rating              0
extracurricular_participation     0
exam_score                        0
dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 16 columns):
 #   Column                         Non-Null Count  Dtype
---  ------                         --------------  -----
 0   student_id                     1000 non-null   object
 1   age                            1000 non-null   int64
 2   gender                         1000 non-null   object
 3   study_hours_per_day            1000 non-null   float64
 4   social_media_hours             1000 non-null   float64
 5   netflix_hours                  1000 non-null   float64
 6   part_time_job                  1000 non-null   object
 7   attendance_percentage          1000 non-null   float64
 8   sleep_hours                    1000 non-null   float64
 9   diet_quality                   1000 non-null   object
 10  exercise_frequency             1000 non-null   int64
 11  parental_education_level       1000 non-null   object
 12  internet_quality               1000 non-null   object
 13  mental_health_rating           1000 non-null   int64
 14  extracurricular_participation  1000 non-null   object
 15  exam_score                     1000 non-null   float64
dtypes: float64(6), int64(3), object(7)
memory usage: 125.1+ KB
```

3.2 Preprocessing - Encode Categorical Variables

In [9]:
```
# Encode Categorical Variables
# First, print out all the unique values entered under parental_education_level.
parental_education_level_values = df['parental_education_level'].unique()
print("\nCategorical Values Under parental_education_level: \n\n    ", parental_edu
```

```
Categorical Values Under parental_education_level:

    ['Master' 'High School' 'Bachelor' 'No Education']
```

In [10]:
```
# Next, encode categorical variables if not already encoded.
if not CATEGORICALS_ENCODED:
```

```
    # Create the mappings for categorical valued and boolean valued variables.
    gender_mapping = {'Male': 0, 'Female': 1, 'Other': 3}
    part_time_job_mapping = {'Yes': 1, 'No': 0}
    diet_quality_mapping = {'Fair': 2, 'Good': 3, 'Poor': 1}
    parental_education_mapping = {'Master': 3, 'High School': 2, 'Bachelor': 1, 'No
    internet_quality_mapping = {'Average': 2, 'Good': 3, 'Poor': 1}
    extracurricular_participation_mapping = {'Yes': 1, 'No': 0}

    # Apply the mapping to each respective categprical variable.
    df['gender'] = df['gender'].map(gender_mapping)
    df['part_time_job'] = df['part_time_job'].map(part_time_job_mapping)
    df['diet_quality'] = df['diet_quality'].map(diet_quality_mapping)
    df['parental_education_level'] = df['parental_education_level'].map(parental_ed
    df['internet_quality'] = df['internet_quality'].map(internet_quality_mapping)
    df['extracurricular_participation'] = df['extracurricular_participation'].map(e

    CATEGORICALS_ENCODED = True

# Preview first 5 rows of datasent to verify encoding of categorical values.
df.head()
df.isnull().sum()
```

Out[10]:

```
student_id                      0
age                             0
gender                          0
study_hours_per_day             0
social_media_hours              0
netflix_hours                   0
part_time_job                   0
attendance_percentage           0
sleep_hours                     0
diet_quality                    0
exercise_frequency              0
parental_education_level        0
internet_quality                0
mental_health_rating            0
extracurricular_participation   0
exam_score                      0
dtype: int64
```

4. New Features (Created)

Since we are studying the correlation between behavioural and demographic charastics and academic productivity, we need to ways to measure productivity in terms of both habits (such as exercise, sleep hours, and studying) in order to make a more accurate analysis on the variability through both exploratory analysis and OLS regression. For this reason, we will create features that reflect the (1) the ROI (return on investment) from the positive habits have and (2) the level of bad habits (in the case offered by this dataset, those habits would be there level screen time).

Feature	Data type	Expected Value	Description
study_roi	Float	Value from 0-1 (exclusive)	How efficiently students use their time when studying
productivity_score	Float	0-100%	How much students are productive - including studying and staying healthy
productivity_roi	Float	Value from 0-1 (exclusive)	Return on Investment, i.e. hoe effective student's productivity score was on there exam
problematic_screen_use_score	Float	Value from 0-1 (exclusive)	Screen time addiction as a binary variable based on hours spent on screen
screen_time_addiction	Categorical	'Low₩n(1-3 hours/day)' / 'Medium₩n(3-5.5 hours/day)' / 'High₩n(5.5-7.5 hours/day)' / 'Very High₩n(7.5+ hours/day)'	Level of screen time addiction student suffers from

4.1 Study ROI

In [11]:
```
# Study ROI (How efficiently students use their time when studying)
# Create a new column for study_roi
# This will be calculated as exam score divided by study hours per day

# Define study_roi as exam score divided by study hours per day
df['study_roi'] = df['exam_score'] / df['study_hours_per_day'].replace(0, 0.0001)

# Calculate the average study_roi
average_study_roi = df['study_roi'].mean()
print("Average Study ROI (Exam Score per Study Hour):", average_study_roi)
print('Average Exam Score:', df['exam_score'].mean())
print('Average Study Hours per Day:', df['study_hours_per_day'].mean())
```

```
Average Study ROI (Exam Score per Study Hour): 5327.431045138574
Average Exam Score: 69.6015
Average Study Hours per Day: 3.5501000000000005
```

4.2 Productivity Score

In [12]:
```
# Productivity Score (How much students are productive - including studying and sta
# Create a new column for productivity_score
# This will be calculated as exam score divided by
```

```
# Normalize study_hours_per_day to fit the scale 1-100
scaler = MinMaxScaler(feature_range=(0, 100))
normalized_study_hours_per_day = scaler.fit_transform(df[['study_hours_per_day']])

# Normalize sleep_hours to fit the scale 1-100
scaler = MinMaxScaler(feature_range=(0, 100))
normalized_sleep_hours = scaler.fit_transform(df[['sleep_hours']])

# Normalize exercise_frequency to fit the scale 1-100
scaler = MinMaxScaler(feature_range=(0, 100))
normalized_exercise_frequency = scaler.fit_transform(df[['exercise_frequency']])

# Normalize diet_quality to fit the scale 1-100
scaler = MinMaxScaler(feature_range=(0, 100))
normalized_diet_quality = scaler.fit_transform(df[['diet_quality']])

# Calculate and save productivity_score score as new colums in the dataframe.
df['productivity_score'] = (normalized_diet_quality.flatten() + normalized_study_ho

# Preview first 10 rows in dataframe to view new feature.
print(df[['productivity_score']].head())
```

```
   productivity_score
0           55.147059
1           75.930191
2           26.030593
3           41.737538
4           46.310241
```

4.3 Productivity ROI

In [13]:
```
# Productivity ROI (Return on investment from students productivity)
# Create a new column for productivity_roi
# This will be calculated as exam score divided by productivity score

# Calculate and save productivity score as new colums in the dataframe.
df['productivity_roi'] = (df['exam_score'] / df['productivity_score']).replace(0, 0

# Calculate the average productivity ROI
average_productivity_roi = df['productivity_roi'].mean()
print("Average Productivity ROI (Exam Score per Study Hour):", average_productivity
print('Average Exam Score:', df['exam_score'].mean())
print('Average Productivity Score:', df['productivity_score'].mean())
```

```
Average Productivity ROI (Exam Score per Study Hour): 1.47381067250944
Average Exam Score: 69.6015
Average Productivity Score: 50.30299875974487
```

4.4 Screen Time Addiction

In [14]:
```
# Screen Time Addiction (New Column, feature)
# Define screen time addiction as a binary variable based on hours spent on screen

# Define screen time addiction variables
```

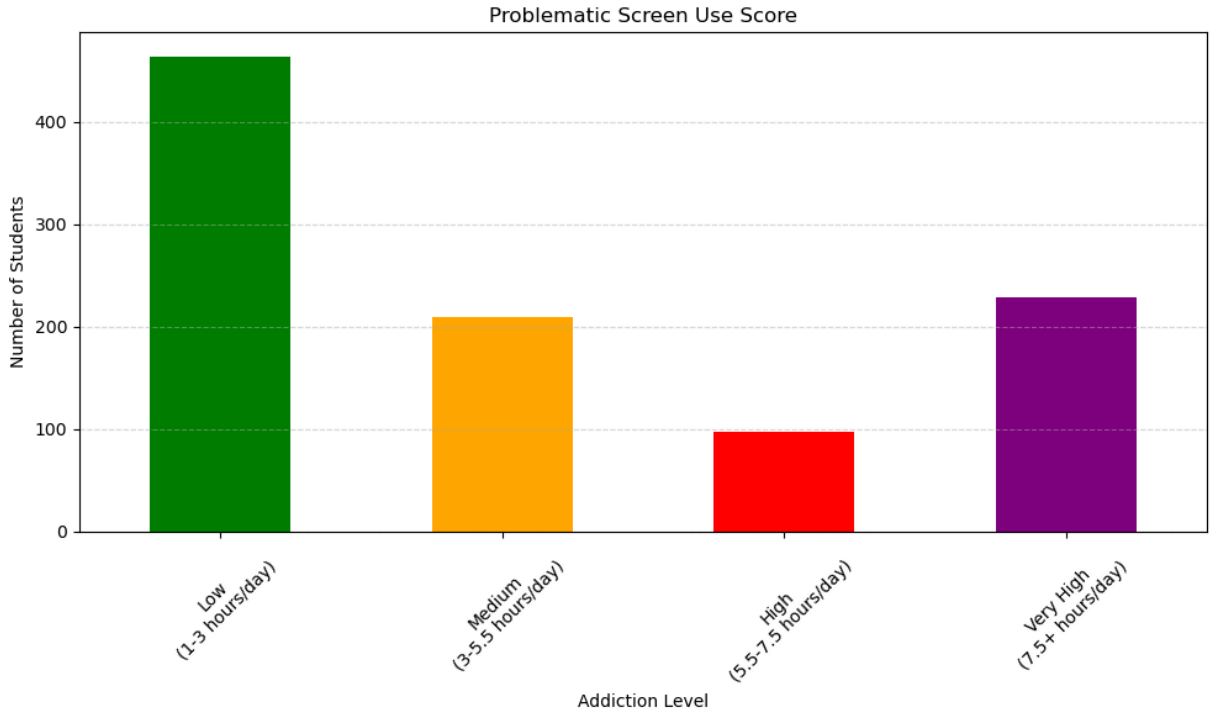
```
def Problematic_Screen_Use_Score(netflix_hours, social_media_hours):
    """
    Measures screen time and the measure problematic usage (addiction) based on Net

    Parameters:
    netflix_hours (float): Daily hours spent on Netflix.
    social_media_hours (float): Daily hours spent on social media.

    Returns:
    float: A score representing problematic screen use. From 0 to 1, 0 (no addictio
    """

# Thresholds
    netflix_threshold = 2  # hours/day
    social_media_threshold = 3  # hours/day
    total_threshold = 5  # total hours/day
    gamma = 1.2 # sensitivity of addiction curve

# Total screen time
    total_screen_time = netflix_hours + social_media_hours

# Penalty if above individual thresholds
    netflix_penalty = 1 + 0.25 * max(0, netflix_hours - netflix_threshold)
    social_media_penalty = 1 + 0.25 * max(0, social_media_hours - social_media_thre

# Combined panalty
    combined_penalty = netflix_penalty * social_media_penalty

# Base addiction score using a logistic function
    base_score = 1 / (1 + math.exp(-gamma * (total_screen_time - total_threshold)))

# Final addiction score with penalties applied, capped at 1
    addiction_score = min(1, base_score * combined_penalty, 1.0)

    return round(addiction_score, 3)

# Add the new problematic screen use score column to the DataFrame
df['problematic_screen_use_score'] = df.apply(lambda row: Problematic_Screen_Use_Sc

# Print the first 5 rows of the new problematic screen use score column
print(df[['problematic_screen_use_score']].head())
```

```
   problematic_screen_use_score
0                         0.038
1                         0.570
2                         0.336
3                         0.576
4                         0.635
```

In [15]:

```
# Calculate the problematic screen use score for each student
df['problematic_screen_use_score'] = df.apply(
    lambda row: Problematic_Screen_Use_Score(row['netflix_hours'], row['social_medi
    axis=1
)

# Categorize problematic screen use score into low, medium, and high addiction grou
```

```python
def categorize_addiction(score):
    if score < 0.3:
        return 'Low\n(1-3 hours/day)'
    elif score < 0.6:
        return 'Medium\n(3-5.5 hours/day)'
    elif score < 0.8:
        return 'High\n(5.5-7.5 hours/day)'
    else:
        return 'Very High\n(7.5+ hours/day)'

df['screen_time_addiction'] = df['problematic_screen_use_score'].apply(categorize_a

# category ordering
category_order = [
    'Low\n(1-3 hours/day)',
    'Medium\n(3-5.5 hours/day)',
    'High\n(5.5-7.5 hours/day)',  # Make sure the catergory order is low to very hi
    'Very High\n(7.5+ hours/day)' # This is the order of categories in the bar char
]

# Count each category
category_counts = df['screen_time_addiction'].value_counts().reindex(category_order

# Plot bar chart
plt.figure(figsize=(10, 6))
category_counts.plot(kind='bar', color=['green', 'orange', 'red', 'purple'])

plt.title('Problematic Screen Use Score')
plt.xlabel('Addiction Level')
plt.ylabel('Number of Students')
plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```



0.0 to 0.3: Low or healthy usage
0.3 to 0.6: Moderate risk
0.6 to 0.8: High risk
0.8 to 1.0: Very high, addicted

5. Exporing Data - Exploratory Data Analysis (EDA)

We explore the distribution and relationships between variables. Below is a correlation heatmap showing the strength of linear relationships with 'exam_score'.

5.1 Descriptive Statistics

In [16]:
```
# Data's central tendency (mean, median), dispersion (standard deviation), and shap
print(df.describe())
```

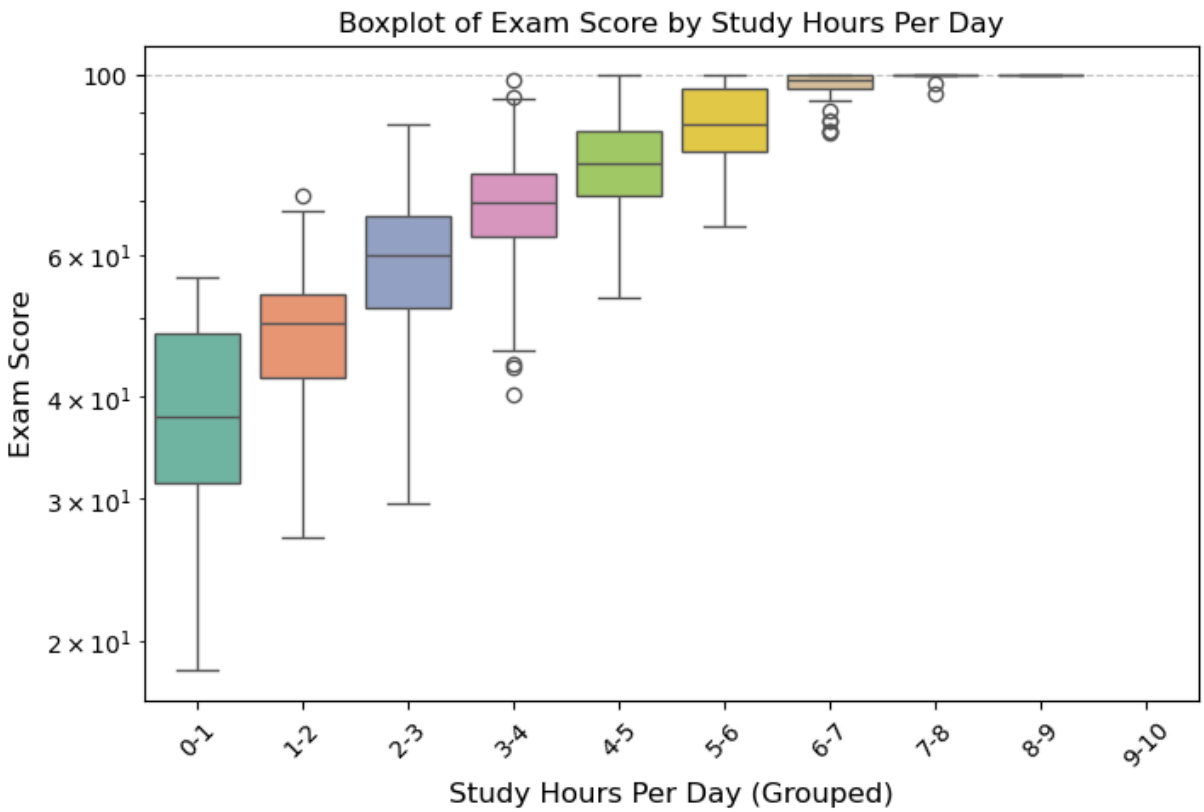
```
             age       gender  study_hours_per_day  social_media_hours  \
count  1000.0000  1000.000000           1000.00000         1000.000000
mean     20.4980     0.607000              3.55010            2.505500
std       2.3081     0.700744              1.46889            1.172422
min      17.0000     0.000000              0.00000            0.000000
25%      18.7500     0.000000              2.60000            1.700000
50%      20.0000     1.000000              3.50000            2.500000
75%      23.0000     1.000000              4.50000            3.300000
max      24.0000     3.000000              8.30000            7.200000

       netflix_hours  part_time_job  attendance_percentage  sleep_hours  \
count    1000.000000    1000.000000            1000.000000  1000.000000
mean        1.819700       0.215000              84.131700     6.470100
std         1.075118       0.411028               9.399246     1.226377
min         0.000000       0.000000              56.000000     3.200000
25%         1.000000       0.000000              78.000000     5.600000
50%         1.800000       0.000000              84.400000     6.500000
75%         2.525000       0.000000              91.025000     7.300000
max         5.400000       1.000000             100.000000    10.000000

       diet_quality  exercise_frequency  parental_education_level  \
count    1000.00000         1000.000000               1000.000000
mean        2.19300            3.042000                  1.635000
std         0.72545            2.025423                  0.865173
min         1.00000            0.000000                  0.000000
25%         2.00000            1.000000                  1.000000
50%         2.00000            3.000000                  2.000000
75%         3.00000            5.000000                  2.000000
max         3.00000            6.000000                  3.000000

       internet_quality  mental_health_rating  extracurricular_participation  \
count       1000.000000           1000.000000                     1000.000000
mean           2.285000              5.438000                        0.318000
std            0.726845              2.847501                        0.465932
min            1.000000              1.000000                        0.000000
25%            2.000000              3.000000                        0.000000
50%            2.000000              5.000000                        0.000000
75%            3.000000              8.000000                        1.000000
max            3.000000             10.000000                        1.000000

        exam_score      study_roi  productivity_score  productivity_roi  \
count  1000.000000    1000.000000         1000.000000       1000.000000
mean     69.601500    5327.431045           50.302999          1.473811
std      16.888564   47984.757018           13.865970          0.505437
min      18.400000      12.048193           13.611800          0.435519
25%      58.475000      16.923280           41.213752          1.147047
50%      70.500000      19.753788           50.348453          1.387697
75%      81.325000      23.873226           60.099959          1.685759
max     100.000000  562000.000000           88.598512          4.231623

       problematic_screen_use_score
count                   1000.000000
mean                       0.433549
std                        0.352440
min                        0.003000
25%                        0.115000
```

```
50%                         0.337500
75%                         0.751250
max                         1.000000
```

5.2 Uncovering Patterns

In [17]:
```
# Boxplot: Exam score by study hours

# Create study hour bins for visualization
bins = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
labels = [f"{bins[i]}-{bins[i+1]}" for i in range(len(bins) - 1)]
df['study_hour_group'] = pd.cut(df['study_hours_per_day'], bins=bins, labels=labels

plt.figure(figsize=(8,5))
sns.boxplot(x='study_hour_group', y='exam_score', data=df, palette="Set2")
plt.xlabel("Study Hours Per Day (Grouped)", fontsize=12)
plt.ylabel("Exam Score", fontsize=12)
plt.xticks(rotation=45)
plt.title("Boxplot of Exam Score by Study Hours Per Day")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout

plt.yscale("log") # Log scale for better spread
plt.gca().yaxis.set_major_formatter(ticker.ScalarFormatter()) # Show plain numbers

plt.show()
```

```
C:\Users\Public\Documents\ESTsoft\CreatorTemp\ipykernel_27404\776925543.py:9: Future
Warning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(x='study_hour_group', y='exam_score', data=df, palette="Set2")
```



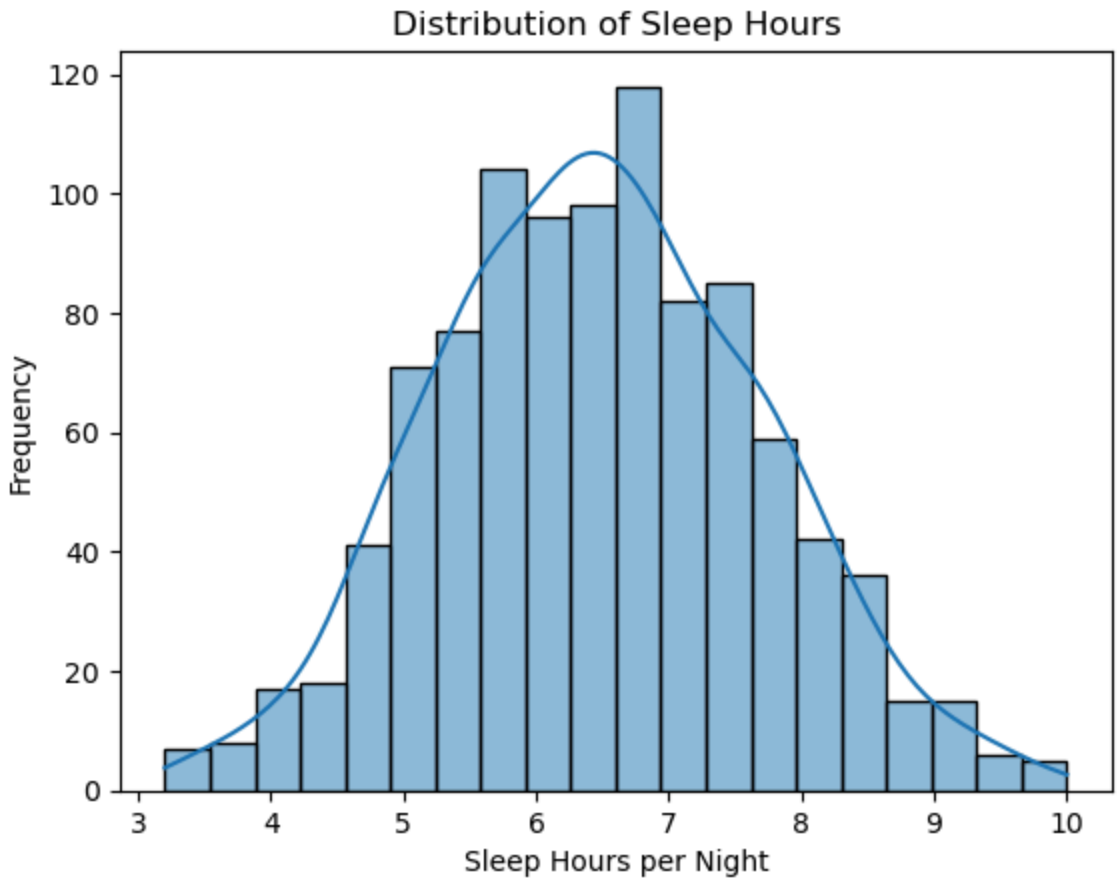
This box plot shows exam scores by study hour range. Students are binned into 1 hour groups.

Key observations:

- There's an upward trend: students who study more hours get higher scores.
- Median score increases across higher study hour groups.
- There's **less variability** in scores for students who study more — more consistent study habits mean more predictable grades.

This plot confirms the positive relationship between study time and productivity, as we found in our regression and correlation analysis.

In [18]:
```
# Histogram: sleep hours

sns.histplot(data=df, x="sleep_hours", kde=True)
plt.title("Distribution of Sleep Hours")
plt.xlabel("Sleep Hours per Night")
plt.ylabel("Frequency")
plt.show()
```



The histogram above shows the distribution of `sleep_hours` in the dataset.

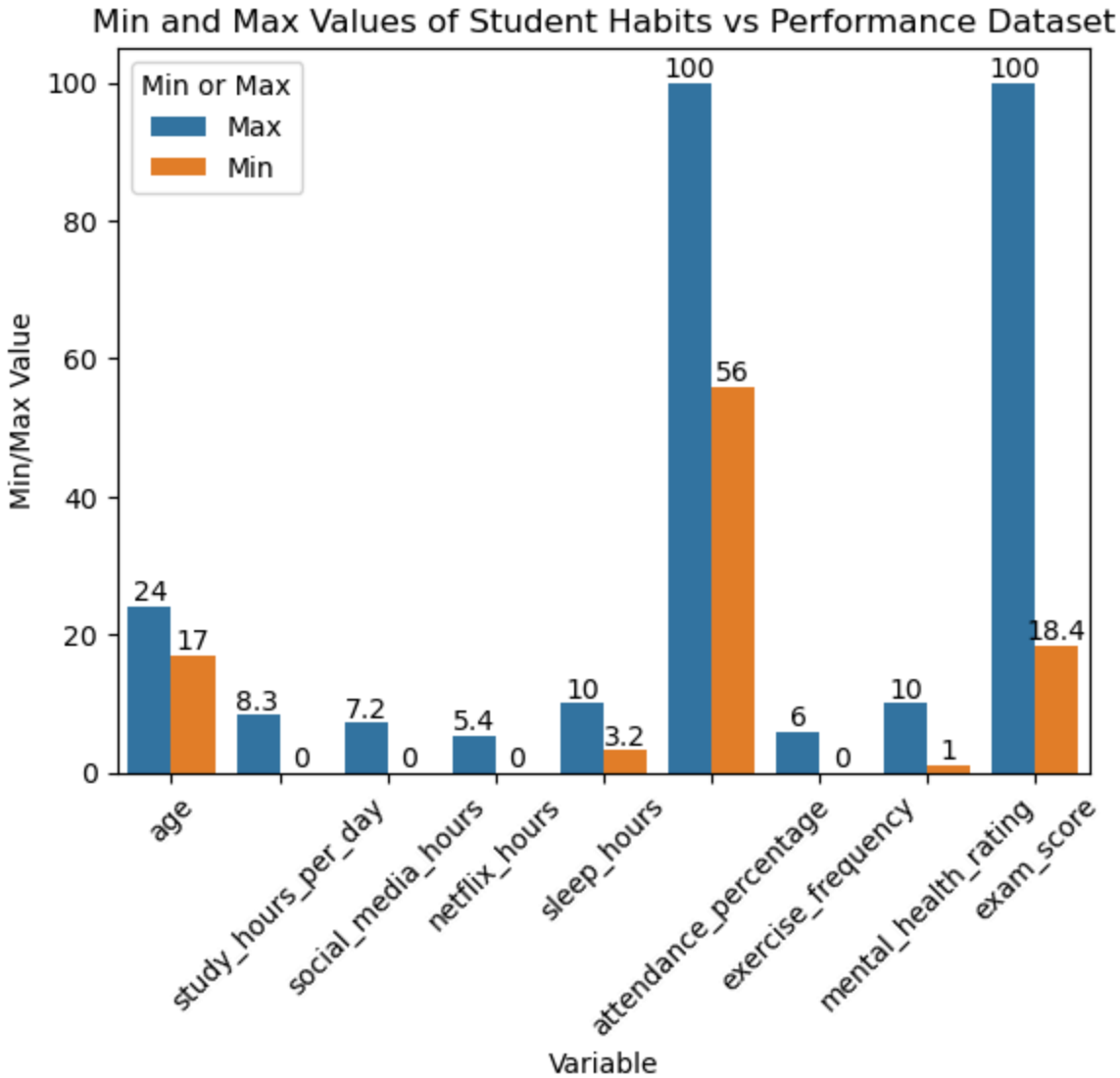
Key observations:

- Most students sleep between **6 to 8 hours** per night, which is the general health recommendation for adults.
- A few students sleep less than 5 hours or more than 9 hours, which could be irregular sleep patterns or lifestyle imbalances.
- The distribution is slightly **right skewed**, with a few students sleeping more than average.

This visualization suggests we should explore whether students with **longer or more consistent sleep** do better academically.

In [19]:
```
# Find max & min values

# Create data frame with min and max values of selected variables.
minMaxDf = pd.DataFrame(columns=['Variable', 'Min/Max Value', 'Min or Max'])

# Add entry for each variable.
for col in ['age', 'study_hours_per_day', 'social_media_hours', 'netflix_hours', 's
    maxData = df[col].max()
    minData = df[col].min()
    maxEntry = pd.DataFrame({'Variable': [col], 'Min/Max Value': [maxData], 'Min or
    minEntry = pd.DataFrame({'Variable': [col], 'Min/Max Value': [minData], 'Min or
    minMaxDf = pd.concat([minMaxDf, maxEntry, minEntry], ignore_index=True)
```

```
# Plot bar graph to show min and max variables.
barplt = sns.barplot(data=minMaxDf, x="Variable", y="Min/Max Value", hue="Min or Ma
plt.xticks(rotation=45)
plt.title("Min and Max Values of Student Habits vs Performance Dataset")

for container in barplt.containers:
    barplt.bar_label(container)
plt.show()
```



5.3 Correlation Analysis

- Identify relationships
- Feature selection
- Avoid Misinterpretation

In [20]:
```
# Calculating correlation between numerical features
correlation_matrix = df.drop(columns=['student_id', 'screen_time_addiction', 'study
```

```
# Display the correlation matrix
print("Correlation Matrix:\n", correlation_matrix)
```

```
Correlation Matrix:
                                   age    gender  study_hours_per_day  \
age                           1.000000 -0.009461             0.003971
gender                       -0.009461  1.000000             0.023329
study_hours_per_day           0.003971  0.023329             1.000000
social_media_hours           -0.009151  0.012503             0.020282
netflix_hours                -0.001174  0.025832            -0.031158
part_time_job                -0.011680 -0.015657            -0.029133
attendance_percentage        -0.026055  0.016514             0.026264
sleep_hours                   0.037482  0.050027            -0.027757
diet_quality                  0.004117 -0.035741             0.033377
exercise_frequency           -0.003836 -0.053949            -0.028701
parental_education_level     -0.040719 -0.018897            -0.000011
internet_quality              0.007799  0.058969             0.014459
mental_health_rating         -0.045101  0.006087            -0.003768
extracurricular_participation -0.004993  0.009118            -0.003264
exam_score                   -0.008907  0.016840             0.825419
study_roi                    -0.058358  0.011199            -0.267616
productivity_score            0.013812 -0.032498             0.314415
productivity_roi             -0.023509  0.043311             0.293905
problematic_screen_use_score -0.021220  0.006699            -0.005694

                                social_media_hours  netflix_hours  \
age                                      -0.009151      -0.001174
gender                                    0.012503       0.025832
study_hours_per_day                       0.020282      -0.031158
social_media_hours                        1.000000       0.011477
netflix_hours                             0.011477       1.000000
part_time_job                             0.021224       0.009207
attendance_percentage                     0.040479      -0.002092
sleep_hours                               0.018236      -0.000935
diet_quality                              0.011344      -0.009885
exercise_frequency                       -0.037319      -0.006448
parental_education_level                  0.021817       0.018715
internet_quality                          0.036805       0.039563
mental_health_rating                      0.001496       0.008034
extracurricular_participation            -0.018597      -0.005125
exam_score                               -0.166733      -0.171779
study_roi                                -0.008483      -0.040828
productivity_score                       -0.002893      -0.020635
productivity_roi                         -0.111815      -0.100922
problematic_screen_use_score              0.683642       0.644009

                                part_time_job  attendance_percentage  \
age                                 -0.011680              -0.026055
gender                              -0.015657               0.016514
study_hours_per_day                 -0.029133               0.026264
social_media_hours                   0.021224               0.040479
netflix_hours                        0.009207              -0.002092
part_time_job                        1.000000              -0.041771
attendance_percentage               -0.041771               1.000000
sleep_hours                          0.001645               0.013756
diet_quality                         0.035266              -0.058621
exercise_frequency                  -0.021679              -0.007857
parental_education_level             0.021041              -0.105619
internet_quality                     0.009130              -0.039903
```

mental_health_rating	0.013539	-0.018745
extracurricular_participation	-0.022841	-0.017778
exam_score	-0.026608	0.089836
study_roi	0.033885	0.002531
productivity_score	0.001108	-0.030266
productivity_roi	-0.009822	0.084257
problematic_screen_use_score	0.033043	0.032734

	sleep_hours	diet_quality	exercise_frequency	\
age	0.037482	0.004117	-0.003836	
gender	0.050027	-0.035741	-0.053949	
study_hours_per_day	-0.027757	0.033377	-0.028701	
social_media_hours	0.018236	0.011344	-0.037319	
netflix_hours	-0.000935	-0.009885	-0.006448	
part_time_job	0.001645	0.035266	-0.021679	
attendance_percentage	0.013756	-0.058621	-0.007857	
sleep_hours	1.000000	-0.034800	0.019769	
diet_quality	-0.034800	1.000000	0.005378	
exercise_frequency	0.019769	0.005378	1.000000	
parental_education_level	-0.007560	0.010279	-0.012379	
internet_quality	0.002045	0.037958	-0.034657	
mental_health_rating	-0.006508	0.027362	-0.000242	
extracurricular_participation	0.027693	-0.030722	-0.005681	
exam_score	0.121683	0.015018	0.160107	
study_roi	-0.012811	-0.010863	0.040894	
productivity_score	0.305583	0.656591	0.609418	
productivity_roi	-0.204826	-0.619580	-0.443285	
problematic_screen_use_score	0.014712	0.008688	-0.021888	

	parental_education_level	internet_quality	\
age	-0.040719	0.007799	
gender	-0.018897	0.058969	
study_hours_per_day	-0.000011	0.014459	
social_media_hours	0.021817	0.036805	
netflix_hours	0.018715	0.039563	
part_time_job	0.021041	0.009130	
attendance_percentage	-0.105619	-0.039903	
sleep_hours	-0.007560	0.002045	
diet_quality	0.010279	0.037958	
exercise_frequency	-0.012379	-0.034657	
parental_education_level	1.000000	0.027100	
internet_quality	0.027100	1.000000	
mental_health_rating	-0.058969	-0.048283	
extracurricular_participation	-0.012242	-0.031420	
exam_score	-0.038813	-0.036298	
study_roi	-0.002013	-0.013827	
productivity_score	-0.003274	0.009009	
productivity_roi	-0.010661	-0.032682	
problematic_screen_use_score	0.016796	0.045338	

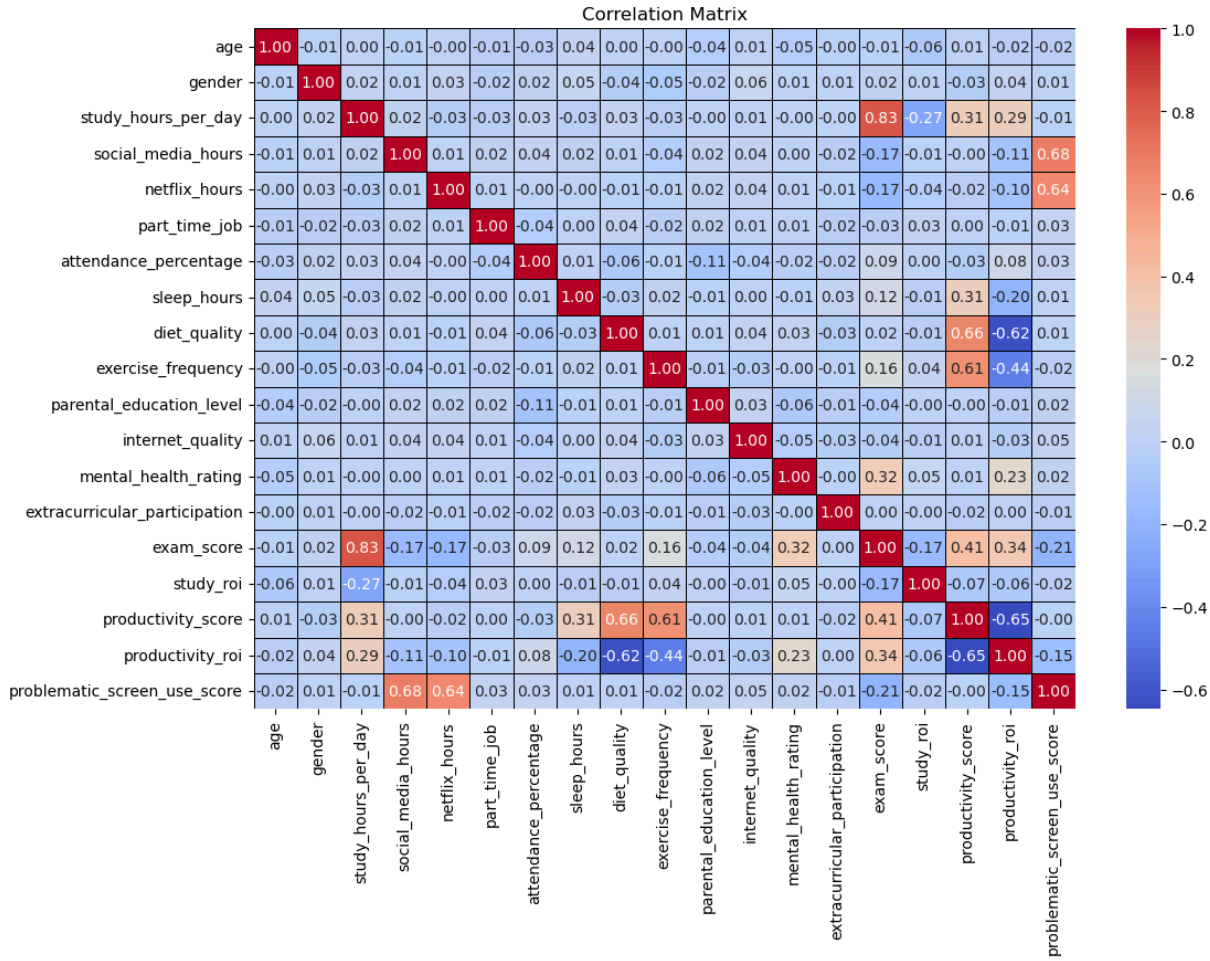
	mental_health_rating	\
age	-0.045101	
gender	0.006087	
study_hours_per_day	-0.003768	
social_media_hours	0.001496	
netflix_hours	0.008034	

```
part_time_job                            0.013539
attendance_percentage                   -0.018745
sleep_hours                             -0.006508
diet_quality                             0.027362
exercise_frequency                      -0.000242
parental_education_level                -0.058969
internet_quality                        -0.048283
mental_health_rating                     1.000000
extracurricular_participation           -0.004741
exam_score                               0.321523
study_roi                                0.049643
productivity_score                       0.014429
productivity_roi                         0.233447
problematic_screen_use_score             0.018532

                              extracurricular_participation  exam_score  \
age                                                -0.004993   -0.008907
gender                                              0.009118    0.016840
study_hours_per_day                                -0.003264    0.825419
social_media_hours                                 -0.018597   -0.166733
netflix_hours                                      -0.005125   -0.171779
part_time_job                                      -0.022841   -0.026608
attendance_percentage                              -0.017778    0.089836
sleep_hours                                         0.027693    0.121683
diet_quality                                       -0.030722    0.015018
exercise_frequency                                 -0.005681    0.160107
parental_education_level                           -0.012242   -0.038813
internet_quality                                   -0.031420   -0.036298
mental_health_rating                               -0.004741    0.321523
extracurricular_participation                       1.000000    0.000881
exam_score                                          0.000881    1.000000
study_roi                                          -0.004067   -0.168535
productivity_score                                 -0.015586    0.410210
productivity_roi                                    0.001322    0.339057
problematic_screen_use_score                       -0.014359   -0.214783

                              study_roi  productivity_score  \
age                           -0.058358            0.013812
gender                         0.011199           -0.032498
study_hours_per_day           -0.267616            0.314415
social_media_hours            -0.008483           -0.002893
netflix_hours                 -0.040828           -0.020635
part_time_job                  0.033885            0.001108
attendance_percentage          0.002531           -0.030266
sleep_hours                   -0.012811            0.305583
diet_quality                  -0.010863            0.656591
exercise_frequency             0.040894            0.609418
parental_education_level      -0.002013           -0.003274
internet_quality              -0.013827            0.009009
mental_health_rating           0.049643            0.014429
extracurricular_participation -0.004067           -0.015586
exam_score                    -0.168535            0.410210
study_roi                      1.000000           -0.071772
productivity_score            -0.071772            1.000000
productivity_roi              -0.056794           -0.647815
problematic_screen_use_score  -0.018901           -0.004673
```

```
                                productivity_roi  problematic_screen_use_score
age                                     -0.023509                     -0.021220
gender                                   0.043311                      0.006699
study_hours_per_day                      0.293905                     -0.005694
social_media_hours                      -0.111815                      0.683642
netflix_hours                           -0.100922                      0.644009
part_time_job                           -0.009822                      0.033043
attendance_percentage                    0.084257                      0.032734
sleep_hours                             -0.204826                      0.014712
diet_quality                            -0.619580                      0.008688
exercise_frequency                      -0.443285                     -0.021888
parental_education_level                -0.010661                      0.016796
internet_quality                        -0.032682                      0.045338
mental_health_rating                     0.233447                      0.018532
extracurricular_participation            0.001322                     -0.014359
exam_score                               0.339057                     -0.214783
study_roi                               -0.056794                     -0.018901
productivity_score                      -0.647815                     -0.004673
productivity_roi                         1.000000                     -0.147624
problematic_screen_use_score            -0.147624                      1.000000
```

In [21]:

```
# Create a heatmap to visualize the correlation matrix
plt.figure(figsize=(12, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f', linewidths=
plt.title('Correlation Matrix')
plt.show()
```



Study hours per day has most strong positive correlation

Social media use and netflix hours has negative correlation

In [22]:
```
# Calculating correlation between study hours per day and exam scores
studyHours_examScores_corr = df['study_hours_per_day'].corr(df['exam_score'])
print("Correlation between Study Hours Per Day and Exam Scores:", studyHours_examSc
```

```
Correlation between Study Hours Per Day and Exam Scores: 0.825418509396044
```

5.4 Analysis Results

Printing out the min and max values of each variables gives us insight into
(1) the demographic of the students that are being analyzed in this dataset and
(2) social implications within this demographic.
In particular we can make the following assumptions and observations:

- More study hours per day result in high academic achievement (exam score)
- Sleep hours per night are mostly distributed around 6.5 hours per night
- The deomgraphic of students in this dataset is 17-24 year olds
- No student got a zero on the exam (the min exam score being 18.4%), and at least one studetn studied no hours- so even student who never studies did not get a zero on their exam.

- There are students who do not watch Netflix and students who do not use social media daily.
- No students attended less than ~half (56 %) of classes in their term/school year.

6. Methodology

6.1 Simple Linear Regression

OLS Regression Model

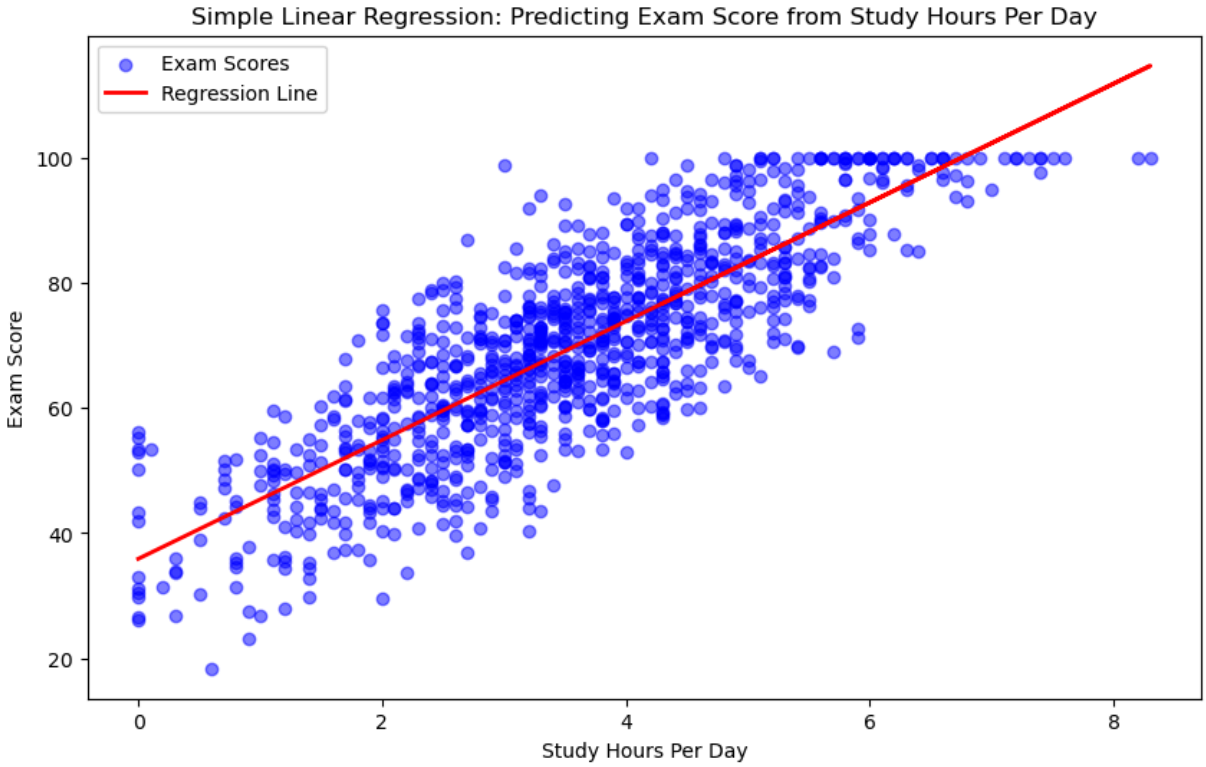
We use Ordinary Least Squares (OLS) regression to identify which variables significantly influence academic performance ('exam_score').

In [23]:
```
# Scatter plot of study hours per day vs exam scores
plt.figure(figsize=(15, 10))
plt.scatter(df['study_hours_per_day'], df['exam_score'], alpha=0.5, color='blue')
plt.xlabel('Study Hours Per Day')
plt.ylabel('Exam Score')
plt.title('Study Hours Per Day vs Exam Score')
plt.xscale("log")
plt.yscale("log")
plt.savefig("scatter_study_hours_exam_scores.png", dpi=300, bbox_inches='tight')
plt.show()
```

In [24]:
```
# Preparing data for simple linear regression
X = df['study_hours_per_day'].values.reshape(-1, 1)
```

```
y = df['exam_score'].values

# Creating and training the model
model = LinearRegression()
model.fit(X, y)

# Making predictions
df['predicted_exam_score'] = model.predict(X)

# Displaying the regression coefficient and intercept
print(f"Regression Coefficient (Slope): {model.coef_[0]}")
print(f"Intercept: {model.intercept_}")

# Plotting the regression line
plt.figure(figsize=(10, 6))
plt.scatter(df['study_hours_per_day'], df['exam_score'], alpha=0.5, color='blue', l
plt.plot(df['study_hours_per_day'], df['predicted_exam_score'], color='red', linewi
plt.xlabel('Study Hours Per Day')
plt.ylabel('Exam Score')
plt.title('Simple Linear Regression: Predicting Exam Score from Study Hours Per Day
plt.legend()
plt.savefig("regression_study_hours_exam_scores.png", dpi=300, bbox_inches='tight')
plt.show()
```

```
Regression Coefficient (Slope): 9.49025040589683
Intercept: 35.91016203402566
```



In [25]:
```
# Define dependent and independent variables
X1 = df[['study_hours_per_day']]
y = df['exam_score']

# Add constant
X = sm.add_constant(X1)
```

```
# OLS regression
model = sm.OLS(y, X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             exam_score   R-squared:                       0.681
Model:                            OLS   Adj. R-squared:                  0.681
Method:                 Least Squares   F-statistic:                     2134.
Date:                Thu, 12 Jun 2025   Prob (F-statistic):          4.60e-250
Time:                        04:44:19   Log-Likelihood:                -3673.3
No. Observations:                1000   AIC:                             7351.
Df Residuals:                     998   BIC:                             7360.
Df Model:                           1
Covariance Type:            nonrobust
==================================================================================
===
                         coef    std err          t      P>|t|      [0.025      0.9
75]
----------------------------------------------------------------------------------
---
const                 35.9102      0.789     45.496      0.000      34.361      37.
459
study_hours_per_day    9.4903      0.205     46.191      0.000       9.087       9.
893
==============================================================================
Omnibus:                        3.233   Durbin-Watson:                   2.012
Prob(Omnibus):                  0.199   Jarque-Bera (JB):                2.727
Skew:                          -0.003   Prob(JB):                        0.256
Kurtosis:                       2.744   Cond. No.                         10.6
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
```

`**68.1% of the variability in students'exam scores can be explained by their study hours alone**`

This indicate that 'Study_Hours_Per_Day' is a strong predictor and study hour fits for prediction model.

6.2 Multiple Linear Regression

In [26]:
```
# Define dependent and independent variables
X = df.drop(columns=['student_id', 'exam_score', 'screen_time_addiction', 'study_ho
y = df['exam_score']
# Add constant
X = sm.add_constant(X)

# OLS regression
model = sm.OLS(y, X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             exam_score   R-squared:                       0.930
Model:                            OLS   Adj. R-squared:                  0.929
Method:                 Least Squares   F-statistic:                     772.6
Date:                Thu, 12 Jun 2025   Prob (F-statistic):               0.00
Time:                        04:44:19   Log-Likelihood:                -2912.3
No. Observations:                1000   AIC:                             5861.
Df Residuals:                     982   BIC:                             5949.
Df Model:                          17
Covariance Type:            nonrobust
==================================================================================
=============
                                 coef    std err          t      P>|t|      [0.02
5      0.975]
----------------------------------------------------------------------------------
-------------
const                           -1.8923      0.110    -17.199      0.000      -2.10
8      -1.676
age                              0.0075      0.062      0.121      0.903      -0.11
4       0.129
gender                           0.0234      0.204      0.114      0.909      -0.37
8       0.425
study_hours_per_day              7.1456      0.411     17.388      0.000       6.33
9       7.952
social_media_hours              -2.0391      0.262     -7.776      0.000      -2.55
4      -1.524
netflix_hours                   -1.7614      0.273     -6.459      0.000      -2.29
7      -1.226
part_time_job                   -0.0107      0.347     -0.031      0.975      -0.69
2       0.671
attendance_percentage            0.1083      0.015      7.039      0.000       0.07
8       0.139
sleep_hours                      0.0045      0.131      0.035      0.972      -0.25
2       0.261
diet_quality                    -5.1030      0.268    -19.009      0.000      -5.63
0      -4.576
exercise_frequency              -0.6790      0.125     -5.417      0.000      -0.92
5      -0.433
parental_education_level        -0.1146      0.166     -0.690      0.490      -0.44
0       0.211
internet_quality                -0.2014      0.197     -1.022      0.307      -0.58
8       0.185
mental_health_rating             1.3689      0.058     23.705      0.000       1.25
6       1.482
extracurricular_participation    0.1600      0.306      0.523      0.601      -0.44
0       0.760
study_roi                     7.681e-06    3.1e-06      2.482      0.013    1.61e-0
6    1.38e-05
productivity_score               0.8388      0.037     22.692      0.000       0.76
6       0.911
productivity_roi                12.7263      0.636     20.019      0.000      11.47
9      13.974
problematic_screen_use_score     0.5458      1.134      0.482      0.630      -1.67
9       2.770
predicted_exam_score            -0.1392      0.054     -2.580      0.010      -0.24
```

```
5      -0.033
==============================================================================
Omnibus:                       61.013   Durbin-Watson:                   1.954
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               93.356
Skew:                          -0.485   Prob(JB):                     5.34e-21
Kurtosis:                       4.139   Cond. No.                     2.69e+20
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
[2] The smallest eigenvalue is 3.22e-29. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

`**P-values**`

Significant predictors that have P-value smaller than 0.05 are

- Study hours per day
- Social media hours
- Netflix hours
- Attendance percentage
- Diet Quality
- Exercise frequency
- Mental health rating
- Study ROI
- Productivity Score
- Productivity ROI

The rest of the variables are not significant with P values higher than 0.05

- Age: 0.903
- Gender: 0.909
- Part-time job: 0.975
- Sleep hours: 0.972
- Parental education level: 0.490
- Internet quality: 0.307
- Extracurricular participation: 0.630
- Problematic_screen_use_score: 0.630

`**F-statistic**`

F-statistic is global test for regression model and test to reject the null.

$$F - statistic = 772.6$$

This model has statistically significant explanatory power.

`**R-squared**`

This Regression model already shows very significant accuracy and predictive power with R-squared of

$$R^2 = 0.930$$

`**Adjusted R-squared**`

$$Adj.\, R^2 = 0.929$$

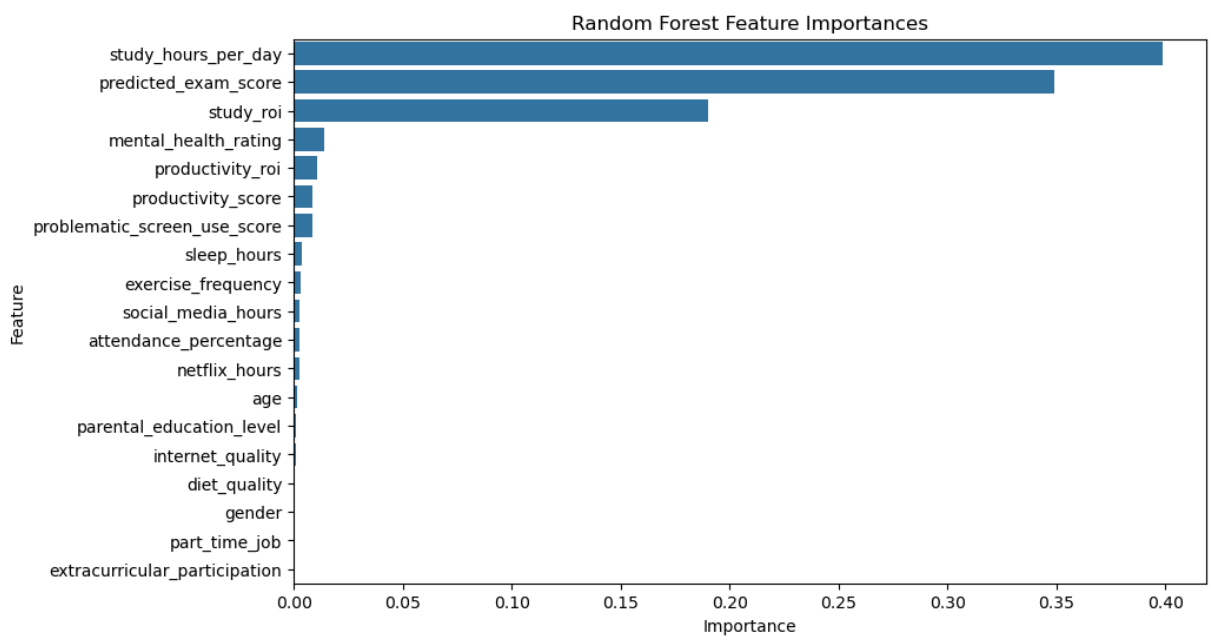
No redundant predictors are included in the model as Adjusted R-squared is nearly the same as R-squared.

6.3 Random Forest Feature Importance

To validate our regression findings, we run a Random Forest model and extract feature importances.
This allows us to see which features contribute most to predicting academic performance, even with non-linear effects.

In [27]:
```
model_rf = RandomForestRegressor(random_state=42)
model_rf.fit(X.drop(columns=['const']), y)

importances = model_rf.feature_importances_
importance_df = pd.DataFrame({'Feature': X.columns[1:], 'Importance': importances})
```

In [28]:
```
# Plot feature importances
plt.figure(figsize=(10, 6))
sns.barplot(x='Importance', y='Feature', data=importance_df)
plt.title('Random Forest Feature Importances')
plt.show()
```



6.4 Choosing the Final Multiple Linear Regression Model

In [29]:
```
# Iterate through each combination of features for the optimal Regression model bas
from itertools import combinations

features = [
    "study_hours_per_day",
    "social_media_hours",
    "netflix_hours",
    "attendance_percentage",
    "diet_quality",
    "exercise_frequency",
    "mental_health_rating",
    "study_roi",
    "productivity_score",
    "productivity_roi"
]

count = 0

for k in range(2, len(features) + 1):
    for combination in combinations(features, k):
        # Define dependent and independent variables
        X = df[list(combination)]
        y = df['exam_score']

        # Add constant
        X = sm.add_constant(X)

        # OLS regression
        model = sm.OLS(y, X).fit()

        # After playing around with the conditions, the most optimal model is narro
        if model.rsquared_adj > 0.85 and model.fvalue > 1700:
            print(combination)
            print("Adj. R-squared: ", model.rsquared_adj, "  |  F-statistic: ", mod
            print("\n=====================================\n")
            count+=1

print(count)
```

```
('study_hours_per_day', 'productivity_score', 'productivity_roi')
Adj. R-squared:  0.8648405048052066   |  F-statistic:  2131.7558724237356

========================================

('study_hours_per_day', 'social_media_hours', 'productivity_score', 'productivity_ro
i')
Adj. R-squared:  0.8735811842887502   |  F-statistic:  1726.8261719082075

========================================

('study_hours_per_day', 'mental_health_rating', 'productivity_score', 'productivity_
roi')
Adj. R-squared:  0.89064872866299   |  F-statistic:  2035.1740636745308

========================================

('study_hours_per_day', 'social_media_hours', 'mental_health_rating', 'productivity_
score', 'productivity_roi')
Adj. R-squared:  0.9035756944895197   |  F-statistic:  1873.291667575287

========================================

('study_hours_per_day', 'netflix_hours', 'mental_health_rating', 'productivity_scor
e', 'productivity_roi')
Adj. R-squared:  0.8986459947154484   |  F-statistic:  1772.5083803551802

========================================

('study_hours_per_day', 'diet_quality', 'mental_health_rating', 'productivity_scor
e', 'productivity_roi')
Adj. R-squared:  0.8996354491961361   |  F-statistic:  1791.9427313699307

========================================

('study_hours_per_day', 'social_media_hours', 'netflix_hours', 'mental_health_ratin
g', 'productivity_score', 'productivity_roi')
Adj. R-squared:  0.9125256556964253   |  F-statistic:  1737.915239354882

========================================

('study_hours_per_day', 'social_media_hours', 'diet_quality', 'mental_health_ratin
g', 'productivity_score', 'productivity_roi')
Adj. R-squared:  0.9137928764166817   |  F-statistic:  1765.8949135430726

========================================

('study_hours_per_day', 'social_media_hours', 'netflix_hours', 'diet_quality', 'ment
al_health_rating', 'productivity_score', 'productivity_roi')
Adj. R-squared:  0.9241532933848976   |  F-statistic:  1739.9005144973423

========================================

9
```

Models with the Highest Adjusted R-value and F-statistic

Model Number	Features	Adjusted R-squared	F-statistic
1	study_hours_per_day, productivity_score, productivity_roi	0.8648	2131.76
2	study_hours_per_day, social_media_hours, productivity_score, productivity_roi	0.8736	1726.83
3	study_hours_per_day, mental_health_rating, productivity_score, productivity_roi	0.8906	2035.17
4	study_hours_per_day, social_media_hours, mental_health_rating, productivity_score, productivity_roi	0.9036	1873.29
5	study_hours_per_day, netflix_hours, mental_health_rating, productivity_score, productivity_roi	0.8986	1772.51
6	study_hours_per_day, diet_quality, mental_health_rating, productivity_score, productivity_roi	0.8996	1791.94
7	study_hours_per_day, social_media_hours, netflix_hours, mental_health_rating, productivity_score, productivity_roi	0.9125	1737.92
8	study_hours_per_day, social_media_hours, diet_quality, mental_health_rating, productivity_score, productivity_roi	0.9138	1765.89
9	study_hours_per_day, social_media_hours, netflix_hours, diet_quality, mental_health_rating, productivity_score, productivity_roi	0.9242	1739.90

Model Picking

Based on these models, it seems the optimal model is #9, because although there are indeed models with significantly higher f-statistics (like model #3) they have lower R-squared indicating that not as much of the variability is explained in those models. Model #9 has the highest R-squared, indicating that most of the variablilty can be explained by the combination of study_hours_per_day, social_media_hours, netflix_hours, diet_quality, mental_health_rating, productivity_score, and productivity_roi. Additionally, the f-statisitc is significantly higher than the initial multiple regression model with all the features- with a large difference of ~1000.

6.4 Final OLS Multiple Regression Model

In [30]:
```
# Conducting final Multiple linear regression with optimal F-statisitic and Adjuste
# Define dependent and independent variables
X = df[['study_hours_per_day', 'social_media_hours', 'netflix_hours', 'diet_quality
y = df['exam_score']
```

```python
# Add constant
X = sm.add_constant(X)

# OLS regression
model = sm.OLS(y, X).fit()
print(combination)
print(model.summary())
```

```
('study_hours_per_day', 'social_media_hours', 'netflix_hours', 'attendance_percentag
e', 'diet_quality', 'exercise_frequency', 'mental_health_rating', 'study_roi', 'prod
uctivity_score', 'productivity_roi')
                            OLS Regression Results
==============================================================================
Dep. Variable:             exam_score   R-squared:                       0.925
Model:                            OLS   Adj. R-squared:                  0.924
Method:                 Least Squares   F-statistic:                     1740.
Date:                Thu, 12 Jun 2025   Prob (F-statistic):               0.00
Time:                        04:44:31   Log-Likelihood:                -2952.0
No. Observations:                1000   AIC:                             5920.
Df Residuals:                     992   BIC:                             5959.
Df Model:                           7
Covariance Type:            nonrobust
==================================================================================
====
                          coef    std err          t      P>|t|      [0.025      0.
975]
----------------------------------------------------------------------------------
----
const                   -0.1098      1.735     -0.063      0.950      -3.515
3.295
study_hours_per_day      6.0289      0.159     37.845      0.000       5.716
6.342
social_media_hours      -1.8474      0.130    -14.195      0.000      -2.103      -
1.592
netflix_hours           -1.6419      0.140    -11.689      0.000      -1.918      -
1.366
diet_quality            -3.5920      0.290    -12.379      0.000      -4.161      -
3.023
mental_health_rating     1.3441      0.059     22.652      0.000       1.228
1.461
productivity_score       0.7311      0.022     32.537      0.000       0.687
0.775
productivity_roi        13.3785      0.652     20.516      0.000      12.099      1
4.658
==============================================================================
Omnibus:                       61.068   Durbin-Watson:                   1.963
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               91.159
Skew:                          -0.494   Prob(JB):                     1.60e-20
Kurtosis:                       4.100   Cond. No.                         659.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
```

`**F-statistic**`

$$F - statistic = 1740$$

F-statistic significantly improved by almost 1000

`**Adjusted R-squared**`

$$Adj.\, R^2 = 0.924$$

Adjusted R-squared slightly got decrease by 0.005

`**Why we choose this model**`

We chose our final OLS regression model based on a combination of statistical power and theoretical interpretability. Although this model has a slightly lower **adjusted R-squared (0.924)** compared to a previous one (0.929), it has a much higher **F-statistic (1740)**, indicating stronger overall model significance.

The F-statistic tests whether our full set of predictors jointly explain a significant portion of the variance in academic performance. A much higher F-value means that, overall, the variables in our final model are statistically strong and not random.

We chose the final model because:

- It has a **high adjusted R-squared**, still explaining over 92% of the variance in scores.
- It passes the **global significance test** with a very high F-statistic.
- All included variables have **theoretical meaning** and **practical interpretation**.

In short, the final model has a good balance of power and robustness and is clear to interpret — for real world application.

Key Insights:

After conducting exploratory data analysis and OLS regression, we identified several important behavioral and demographic factors that influence academic productivity among university students.

- **Study Hours per Day** is the most influential predictor of academic performance. Students who dedicate more time to studying consistently score higher on exams, as confirmed by both correlation and regression results.

- **Mental health rating**, and **Exercise Frequency** shows a positive relationship with exam scores. Students who exercise and has higher mental health rating perform better than those with significantly low mental health rating or irregular exercise schedules.

- **Attendance Percentage** somehow correlates with higher exam scores. Regular class attendance appears to be a reliable proxy for student engagement and academic success.
- **Diet Quality** were not statistically significant predictors in our final regression model, but may still play indirect roles through mental health and energy levels.
- New features such as **Study ROI**, **Productivity Score**, **Productivity ROI** has significant statistical value and are good predictors, as it has p-value < 0.05 and strongly correlated with exam score.

These insights suggest that academic productivity is most strongly tied to consistent behavioral routines like studying, sleeping well, eating well, and attending classes, rather than demographic or lifestyle factors alone.

7. Results

After generating multiple regression models for all feature combinations, we found the most optimal model that is the best at explaining the variability of student academic performance. The models has the following statistics:

Predictors: features study_hours_per_day, social_media_hours, netflix_hours, diet_quality, mental_health_rating, productivity_score, productivity_roi

$$F - statistic = 1739.90$$

$$Adj.\, R^2 = 0.9242$$

(P-values of all predictor, except the constant, are 0.000)

This model has a significantly higher R-squared (~ 0.4 difference) than the linear regression model and a significantly higher F-statistic than the previous multiple regression model (~1000 difference). Thus, this model is the optimal model for explaining the variability of a student's academic performance (i.e. the variability of the exam score). The combination of the features study_hours_per_day, social_media_hours, netflix_hours, diet_quality, mental_health_rating, productivity_score and productivity_roi, do a significantly better job at explaining.

8. Conclusion

Our analysis aimed to identify the most influential behavioural and demographic factors affecting academic performance (exam score). Using Exportatory Data Analysis (EDA), OLS regression, and Random Forest feature importance, we found that:

- **Study hours per day**, **Mental health rating**, and **attendance percentage** were the strongest predictors of higher exam scores.
- Also following lists were strong predictors as well
  - Social media hours
  - Netflix hours
  - Diet Quality
  - Exercise frequency
  - Study ROI (Created)
  - Productivity Score (Created)
  - Productivity ROI (Created)

Including new features

The rest of the variables are not significant with P values higher than 0.05

- **Age**, **Gender**, **Part-time job** and **Sleep hours** had very weak or inconsistent relationships with academic performance.

- Demographic variables like **Internet quality** and **parental education** showed mild effects.

Limitations:

- Self-reported data may introduce bias.
- The dataset lacks geographic and institutional diversity.
- Academic achivement is measured only by exam scores.

Future Research:

- Use multi-semester or GPA data for stronger performance metrics.
- Explore interaction effects (e.g., does social media affect low vs. high sleepers differently?).
- Integrate mental health metrics more deeply into predictive modeling.

Our findings can help students and educators prioritize lifestyle habits that statistically support better academic outcomes.